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Summary

Final-year Instrumentation and Control Engineering student at NIT Tiruchirappalli with controls internships at EPFL and IIT Madras. Experienced in system modeling, controller design, state estimation and ROS 2 autonomy, tested in simulation and on hardware.

Education

National Institute of Technology Tiruchirappalli Expected May 2027
B.Tech in Instrumentation and Control Engineering, Minor in Computer Applications CGPA: 8.5/10

- **Related Coursework:** Control Systems, Pattern Recognition, Signal Processing, IoT, Fuzzy Logic

Indian Institute of Technology Madras Expected May 2027
BS in Data Science and Applications CGPA: 8.61/10

- **Related Coursework:** Statistics, Data Structures and Algorithms, DBMS, Modern Application Development

Skills

- **Languages and Tools:** Python, C, C++, MATLAB, Bash, Linux, Git, Docker, PostgreSQL
- **Control and Robotics:** PID, LQR, State Estimation (EKF), Kinematics and Dynamics, ROS 2, Nav2, SLAM
- **Machine Learning and Vision:** TensorFlow, NumPy, OpenCV, YOLO
- **Simulation and CAD:** MuJoCo, Gazebo, Stonefish, Simulink, SolidWorks, Onshape, COMSOL
- **Hardware:** ESP32 (ESP-IDF), Arduino, Raspberry Pi, NVIDIA Jetson, Intel RealSense, RPLIDAR, 3D Printing
- **Miscellaneous:** Self-hosted Linux home server, KVM GPU passthrough, local LLM hosting, web development

Experience

Controls Research Intern | Biorobotics Laboratory (BioRob), EPFL June 2026 – Aug 2026
ZBot: Simulating a Bio-Inspired Zebrafish-Larva Robot | Prof. Auke Ijspeert | Report Link Lausanne, Switzerland

- Ported ZBot, an 80 cm zebrafish-larva robot (*Science Robotics*, 2026), from Webots to **FARMSIM**, the lab's **MuJoCo-based neuromusculoskeletal simulator**, and implemented a custom **CPG controller** in Python
- Diagnosed that the simulator applied buoyancy at the center of mass, giving zero righting torque, and corrected it with a **self-derived tetrahedral decomposition** of each body's submerged collision geometry
- Validated the method against closed-form hydrostatics, matching settling depth to 0.26 mm across 12 cases and the densities at which a floating cube capsizes to 0.2 kg/m^3
- Optimized it in Cython with mesh caching, bounding-sphere rejection and 20-step interpolation, giving the robot a 0.5 N m righting moment at 60° roll while running **3.2× faster than real time**

Controls Research Intern | NIT Tiruchirappalli Dec 2024 – Present
Controller Implementation in Rotary Inverted Pendulum | Dr. D. Ezhilarasi | Report Link Tiruchirappalli, India

- Derived the **Euler-Lagrange model** of a laboratory Quanser rotary inverted pendulum, an unstable underactuated system with 1 input and 4 states, and linearized it about the upright equilibrium
- Designed **pole-placement and LQR** balance controllers in MATLAB and Simulink, with LQR cutting arm overshoot from **12.1% to under 0.1%** on a 20° square-wave setpoint in simulation
- Deployed both on the Quanser hardware through **QUARC real-time control**, with LQR balancing the pendulum while holding a 20° arm setpoint to ±4° steady-state error with 6 s settling
- Extended the stable arm range to ±45°, where both fixed-gain controllers failed, by **gain scheduling** a softer LQR gain at large arm angles

Controls Intern | R2D2 Lab, IIT Madras May 2025 – July 2025
Push-Rim Torque Sensing and Cruise Control for Wheelchair | Thryv Mobility Chennai, India

- Built an **ESP32 data logger in ESP-IDF** that records the torque a user applies to the wheelchair push rims, sensed by strain gauges through an ADS1115 ADC, as the input for cruise control
- Calibrated the strain gauges against known loads from 0.8 to 14 kg in both directions to remove **bias and slope**
- Conducted push trials on **40 subjects** of varying physiques, recording sustained and impulsive push-rim torque
- Formulated a **state-space model** of the wheelchair dynamics and implemented a **cruise control algorithm** on moving-average-filtered torque input, verified on a bench rig as a proof of concept

Projects

ROVIO: 8-Thruster 6-DOF Underwater ROV, Simulation & Autonomy | Report Link Dec 2024 – Present

- Built the simulation of the RMI club ROV in **Stonefish** with **ROS 2** sensor, thruster and teleoperation interfaces as the testbed for its autonomy stack, and published an open-source reef scene generator
- Designed a **6-DOF cascade controller** with drag feedforward from a derived Newtonian model of thruster forces, mass and inertia, holding depth to 0.4 cm and tracking velocities within 10% in water currents
- Closed the control loops on a custom **13-state extended Kalman filter (EKF)** by fusing IMU, pressure, and visual fixes, with estimated depth having RMS error of 0.1 cm
- Localized the vehicle from **AprilTag** boards with a **Levenberg-Marquardt** nonlinear least-squares fit and an IMU tilt prior, reaching 5–12 cm position and under 1° yaw error
- Implemented autonomous **waypoint navigation** and terrain-following surveys, used to **map 50 of 53 corals** with a YOLO11, ByteTrack and multi-view triangulation pipeline

Trident: Differential-Drive Pick and Place Robot, Design & Simulation | Report Link Sep 2025 – Dec 2025

- Built a 0.64 kg **mobile robot** for a **WorldSkills** line-following and vision-based **coin-sorting task**, modeling it in SolidWorks, 3D printing the chassis and wiring the electronics in 15 days
- Designed a **single-servo gripper** with two geared **parabolic jaws** that cages a 40 mm coin and slides it along the floor, with **ejector** flaps that push the coin clear as the jaws open
- Fused wheel speed and gyroscope yaw rate in an **EKF** for odometry, staying within 30 mm of ground truth over an 18 m mission in simulation
- Modeled the robot in Gazebo and ROS 2 with an emulated 8-channel infrared array, and wrote a **PD line follower** that tracks within 6 mm where the depth camera cannot see the floor
- Integrated RTAB-Map **visual SLAM** and **Nav2** with a custom docking and mission state machine, delivering **36 of 36 coins** in 12 simulated missions with obstacles at 10 mm RMS localization error

Mechanical Watch Going Train with Swiss Lever Escapement | Report Link Aug 2026 – Sep 2026

- Modeled the timekeeping mechanism of a mechanical watch as an individual project, a fully parametric 13-part SolidWorks **going train** with a **Swiss lever escapement** beating at 28,800 beats per hour
- Constructed **NIHS 20-30 standard ogival** gear teeth parametrically from tangency relations and set center-distance tolerances with a 4,000-sample **Monte Carlo** contact study
- Calculated bending, Hertz contact, and fatigue stresses on every load-carrying part under a **5,000 g shock**, and sized the mainspring and barrel for a **40-hour power reserve**
- Simulated the complete train in **SolidWorks Motion** with impact-based contact, measuring a balance period of **0.2498 s against the 0.2500 s** design value
- Proposed manufacturing routes of **wire EDM, LIGA and silicon etching** for the escapement with functional tolerances down to ±0.005 mm

Awards and Achievements

- Won the **district round** of the **WorldSkills competition** with Trident, a self-built pick-and-place robot
- Achieved a **perfect score** in the **CSWA** SolidWorks Certification exam in the modeling and assembly category
- Won **1st** and **2nd** in the **2024 District-level Toastmasters** Humorous Speech and Speech Evaluation Contests
- Won **Gold** and **Bronze** in **2018 Syloth Silambam State-level** tournament and trained a team of 35 who won **7 Gold**, **13 Silver** and **17 Bronze**
- Won **Gold** in Photography, Silver in Group Musical and Bronze in Solo Instrumental at the 2018 State-level Arts competition of Madurai Kamaraj University

Leadership

Technical Head | Robotics and Machine Intelligence (RMI), NIT Tiruchirappalli July 2026 – Present

- Manage **4 concurrent projects** in the **46-member** official robotics club of NIT Tiruchirappalli, mentoring members in simulation, controls, and computer vision
- Conducted Techids, a robotics workshop for government school children, since joining as a member in July 2024

Secretary | Toastmasters NITT Chapter July 2024 – Apr 2025

- Coordinated the core team and administrative activities of the official Toastmasters club of NIT Tiruchirappalli
- Delivered speeches and evaluations at public speaking meetings and contests as a member since March 2024